Quiz 2

1) Which aspect is not included under 'Subsystem Decomposition'?
A) Layers vs Partitions
B) Coherence & Coupling
C) Identification of Parallelism
D) Architectural Style

2) What does the Nonfunctional Requirements analysis model primarily influence in system design?
A) Hardware/Software Mapping
B) Boundary Conditions
C) Definition of Design Goals
D) Software Control

3) The Object model in system design analysis impacts which of the following?
A) Nonfunctional Requirements
B) Identification of Concurrency and Global Resource Handling
C) Hardware/Software Mapping and Persistent Data Management
D) Boundary Conditions

4) How does an API relate to a subsystem interface?
A) An API is the same as a subsystem interface
B) An API is a refinement of the subsystem interface for internal use within the subsystem
C) An API is the specification of the subsystem interface in a specific programming language
D) An API is not related to subsystem interfaces

5) Which of the following best describes low coupling in a system?
A) A change in one subsystem will have a high impact on other subsystems.
B) A change in one subsystem does not affect any other subsystem.
C) Subsystems have many dependencies and associations.
D) Subsystems perform different tasks but are still closely related.

6) What is a defining characteristic of a layer in a subsystem?
A) It depends on services from higher layers.
B) It provides services to partitions on different layers.
C) It only depends on services from lower layers.
D) It has extensive knowledge of higher layers.

7) How does the call operation differ between closed and open architectures?
A) In a closed architecture, layers can call operations only from the layer directly below, whereas in an open architecture, layers can call operations from any layer below.
B) In a closed architecture, layers can call operations from any layer, whereas in an open architecture, layers can only call operations from the layer directly below.
C) Both architectures allow layers to call operations from any layer below.
D) Both architectures restrict layers to call operations only from the layer directly below.

8) Which design goal ensures that the client/server architecture can handle an increasing number of client requests?
A) High Performance
B) Scalability
C) Flexibility
D) Reliability

9) When considering the mapping of control objects to a processor, what is an important factor to consider?
A) The color scheme of the control objects
B) The computational rate demand on a single processor
C) The encryption level of control objects D) The compatibility with mobile devices

10) What is the primary use of deployment diagrams in system design?
A) To illustrate the user interface design
B) To show a system design after design decisions have been made
C) To depict the dynamic aspects of systems
D) To outline the software development process

11) For which scenario is a database recommended as a mechanism for persistent objects?
A) When the data are only read once and never written to.

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B) If the data are used by concurrent writers and readers.
C) If the data are to be archived for long-term storage.
D) When the data are to be stored temporarily and deleted after use.

12) What does a capability associate in the context of system security?
A) A user's level of expertise
B) A class's methods to its performance metrics
C) A (class, operation) pair with an actor D) A set of all possible permissions available to the system

13) What is the primary goal of the reuse activity in object design?
A) To use inheritance to create new classes from scratch
B) To identify and utilize existing solutions, such as off-the-shelf components and design patterns
C) To discard old components and develop newer versions
D) To eliminate the need for inheritance in design

14) When discovering inheritance associations, what is specialization?
A) Creating a new class without any inheritance relationship
B) Identifying unique features in a class to differentiate it from its superclass
C) The discovery of an inheritance relationship where the super class is discovered first
D) The conversion of concrete classes into more abstract entities

15) What is the main difference between inheritance and delegation as described in the slide?
A) Inheritance involves creating entirely new operations, whereas delegation involves modifying existing operations.
B) Inheritance involves extending a base class with new operations or overwriting existing ones, while delegation involves catching an operation and sending it to another object.
C) Inheritance is a pattern used exclusively in Java, while delegation is used in C++.
D) Inheritance is about removing operations from a base class, whereas delegation is about

16) What is a benefit of delegation over inheritance?
A) It is more efficient since objects are encapsulated.
B) It allows an object to be replaced at runtime, provided it is the same type.
C) It exposes a subclass to the details of its parent class.
D) It requires recompilation of the subclass whenever the parent class changes.

17) What does strict inheritance mean in Java?
A) The subclass cannot add any new methods or properties.
B) The subclass can add new methods to the superclass but cannot overwrite them.
C) The subclass must rewrite all methods from the superclass.
D) The superclass can override methods in the subclass.

18) How can you prevent a method from being overwritten in Java?
A) By declaring the method as static.
B) By prefixing the method with the keyword final.
C) By omitting the method from the subclass.
D) By declaring the method as private in the superclass.

19) What defines a framework in software development?
A) A complete, ready-to-use software application.
B) A set of coding guidelines for a specific programming language.
C) A partial application that can be specialized to produce custom applications.
D) A programming tool used for debugging applications.

20) What is contraction in the context of object-oriented programming?
A) A method of optimizing code by reducing its length.
B) Implementations of methods in the superclass being overwritten with empty bodies in the subclass.
C) The process of removing unnecessary classes from a hierarchy.
D) Reducing the visibility of methods from public to private.